

Unit Outline

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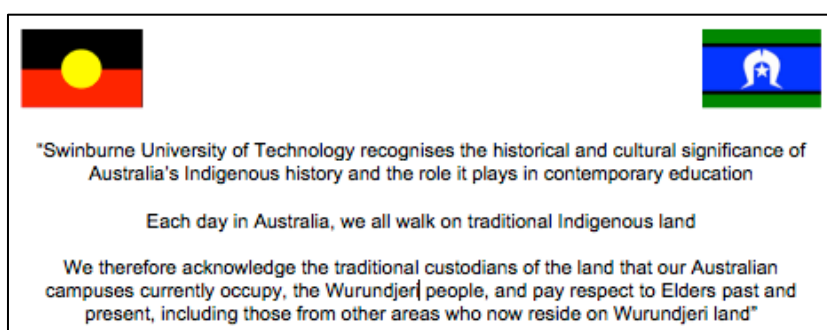
TNE10006

Networks and Switching

Semester 1 2025

Please read this Unit Outline carefully. It includes:

- PART A** Unit summary
- PART B** Your Unit in more detail
- PART C** Further information



PART A: Unit Summary

Unit Code(s)	TNE10006
Unit Title	Networks and Switching
Duration	One Semester or equivalent
Total Contact Hours	60 hours
Requisites:	
Pre-requisites	
Co-requisites	
Concurrent pre-requisites	
Anti-requisites	TNE10003 LAN Principles TNE20001 IP Technologies
Assumed knowledge	
Credit Points	12.5
Campus/Location	Hawthorn
Mode of Delivery	Online Synchronous/Asynchronous and On-campus
Assessment Summary	In-Semester Assessment – 30% Final Skills Assessment – 20% – Hurdle: 50% Final Concepts Assessments – 50% – Hurdle: 40%

Aims

This unit of study aims to introduce you to the field of data networking and to provide you with the necessary skills to design and successfully deploy a small network.

Unit Learning Outcomes

Students who successfully complete this unit can:

1. Use the OSI Model to discuss network design and infrastructure (K3, S3)
2. Design an addressing scheme to subnet a network to given specifications (K2, K3, S3)
3. Describe the processes of LAN switching and how VLANs are used in a switched network (K3)
4. Use best practice to design secure corporate networks and deploy security to practical networking environments (K3, S1, S3)
5. Describe how the Spanning Tree Protocol eliminates loops in multi-switched LANs (K3)
6. Construct a simple network of PCs, switches and a router, and troubleshoot problems in those networks (K3, S1, S2, S3)
7. Generate documentation for laboratory work and assignments (A2)

Graduate Attributes

The Swinburne Graduate Attributes describe the capability of our graduates to use knowledge, skills and behaviours to contribute to society meaningfully and positively. They include professional, self-directed learning and future-ready skills.

This unit contributes to the development of the following Swinburne Graduate Attributes:

- GA5 Digital Literacies - Information Literacy

Other graduate attributes may be practised in the unit but are not formally taught as part of the unit content, nor incorporated within formal assessment.

Content

- Networking Fundamentals
- OSI Network Layers
- Ethernet fundamentals, technologies and switching
- TCP/IP and addressing
- Subnets and subnetting (VLSM and CIDR)
- Switching concepts
- Virtual LANs
- Inter-VLAN Static routing
- Network and Secure Network Design
- EtherChannel Bundling
- Spanning Tree Protocol
- Wireless LANs
- Basic switch troubleshooting History of Unix

PART B: Your Unit in more detail

Unit Improvements

Feedback provided by previous students through the Student Survey has resulted in improvements that have been made to this unit. Recent improvements include:

- Web-based lecture content to complement pre-recorded lectures
- Updated Final Skills Assessment Format
- Additional Revision Resources

Unit Teaching Staff

Name	Role	Room	Email	Consultation Times
Karina Cereceda	Unit Convenor Lecturer	EN607a	ncerecedacastellani@swin.edu.au	Fri 08:30 – 10:30 AM or by appointment
Fahmida Prema	Unit Academic Staff	EN607a	fprema@swin.edu.au	Tue 2:30 – 4:30 PM
Canvas Staff List	Lab Instructors		Contact via Canvas	

Learning and Teaching Structure

Category	Activity	Total Hours	Hours per Week	Teaching Period Weeks
Online	Lectures	24 hours	2 hours	Weeks 1 to 12
In person and Live online	Lecture Workshops	24 hours	2 hours	Weeks 1 to 12
In Person	Labs	36 hours	3 hours	Weeks 1 to 12
Online	NetAcademy Program	30 hours	2.5 hours	Weeks 1 to 12
Online	Self-paced learning activities	36 hours	3 hours	Weeks 1 to 12

Week by Week Schedule

Week	Lecture Topics	Laboratory Topics and Activities	Graded Tasks and Assessments
1	Unit Overview Network Layer Modules	Lab Journal Introduction to ATC Labs Introduction to Packet Tracer Lab SU-1a and Lab SU-2a	
2	Link Layer Ethernet and Switching Concepts	VLANs and LAN Security Switch Security and Remote Access Lab SU-3a	
3	VLANs and LAN Security Trunking	VLANs and LAN Security Trunking Lab SU-4a	
4	Network Layer – IPv4	Inter-VLAN routing Lab SU-5a	
5	IPv4 Subnetting – VLSM	Inter-VLAN routing Lab SU-5b	Group Lab Activity 1 Starts
6	IPv4 Subnetting – VLSM	Subnetting Exercises Troubleshooting Lab SU-6a	Group Lab Activity 1 Continues
7	Inter-VLAN Routing LAN Design	Sample Mid-Sem Skills Assessment Live Demonstration	Group Lab Activity 1 Submission
8	Spanning Tree Protocol	Mid-Term Assessments	VLSM Test Mid-Sem Skills 4 x NetAcademy Checkpoint Exams (Listed on Canvas)
9	EtherChannel Bundling Wireless Concepts	Spanning-Tree Protocol Lab SU-7a and Lab SU-8a	
10	IPv6	EtherChannels Lab SU-9a and Lab SU-9b	
11	Transport Layer – TCP and UDP	Sample Final Skills Assessment	Group Lab Activity 2 Starts
12	Transport Layer – TCP and UDP	Sample Final Skills Assessment Live Demonstration	Group Lab Activity 2 Submission 4 x NetAcademy Checkpoint Exams (Listed on Canvas)

Assessment

a) Assessment overview

Tasks and Details	Individual or Group	Weighting	Mapped Unit Learning Outcomes	Mapped Graduate Attributes	Assessment Due Date
1. NetAcademy Checkpoint Exams	Individual	1%	1, 2	GA5	Refer to weekly schedule
2. Group Lab Activities	Group	15%	1, 2, 3, 4, 5, 6, 7	GA5	Refer to weekly schedule
2. VLSM Test	Individual	5%	2	GA5	Refer to weekly schedule
4. Mid-Semester Skills Assessment	Individual	5%	2, 3, 6, 7	GA5	Refer to weekly schedule
5. NetAcademy CCNA1 Final Exam	Individual	4%	1, 2	GA5	Final Assessment Period
6. Final Skills Assessment	Individual	20%	2, 3, 6, 7	GA5	Final Assessment Period
7. Final Written Assessment	Individual	50%	1, 2, 3, 4, 5	GA5	Final Assessment Period

Assessment Requirements	Details
b) Use of generative AI (genAI) in this unit	The valid use of genAI in this unit is as follows: Assignments and assessments in this unit should be completed entirely without genAI assistance.
c) Hurdle requirements	To pass this Unit, you must: - achieve an overall mark for the unit of 50% or more, and - obtain at least 40% in the Final Written Assessments, and - must pass the Final Skills Assessment.
d) Final assessment period	If the unit you are enrolled in has a final assessment (including invigilated exams), you will be expected to be available for the entire final assessment period including any Special Exam period and Resit dates. The Final Written Assessment is an invigilated, closed-book assessment and calculators are not allowed. The Final Skills Assessment is an invigilated assessment, you can only use your hand-written lab notes for reference and calculators are not allowed.

e) Submission requirements	Group Lab Activities are submitted online through the Canvas assessment submission system which integrates with the Turnitin . Please ensure you keep a copy of all assessments that are submitted through Canvas.
f) Extensions and late submissions	If you were unable to attend an on-campus or online assessment due to serious and unavoidable circumstances outside of your control, you may be able to apply for special consideration (SPC). Please refer to Part C of this document for more information. Students who completed the assessment on the original date will not be granted SPC. Therefore, if you are unfit to complete an assessment, you are advised to not attempt it and apply for SPC. Late Submissions - Unless an extension has been approved, late submissions will result in a penalty. You will be penalised 10% of your achieved mark for each working day the task is late, up to a maximum of 5 working days. After 5 working days, a zero result will be recorded.
g) Referencing	To avoid breaching academic integrity, you are required to provide references whenever you include information from other sources in your work and acknowledge when you have used Artificial Intelligence (AI) tools (such as ChatGPT). Further details regarding academic integrity are available in Section C of this document. Referencing conventions required for this unit are: http://www.ieee.org/documents/ieeecitationref.pdf Helpful information on referencing can be found at http://www.swinburne.edu.au/library/referencing/
h) Groupwork guidelines	One of the main outcomes of Group Lab Activities is learning how to work in a group efficiently and collaboratively. For this reason, teaching staff do not interfere nor manage teamwork, this is the responsibility of the group members. It will be assumed that the document submitted by the group is the result of a collaborative effort and the mark will apply to all members of the group without distinction. All group members must be satisfied that the work has been correctly submitted. Any penalties for late submission will generally apply to all group members, not just the person who submitted.

Required Textbook(s)

None Required

Recommended Reading Materials

Swinburne Library has a large collection of resources. Listed below are some references that will provide valuable supplementary information to this unit. It is also recommended that you explore other sources to broaden your understanding.

- CCNAv7 Introduction to Networks Companion Guide 2021
- Switching, Routing, and Wireless Essentials Companion Guide (CCNAv7) 2021

PART C: FURTHER INFORMATION



For further information on any of these topics, refer to Swinburne's Student webpage <http://www.swinburne.edu.au/student/>

Student behaviour and wellbeing

All students are expected to: act with integrity, honesty and fairness; be inclusive, ethical and respectful of others; and appropriately use University resources, information, equipment and facilities. All students are expected to contribute to creating a work and study environment that is safe and free from bullying, violence, discrimination, sexual harassment, vilification and other forms of unacceptable behaviour.

The [Student Charter](#) describes what students can reasonably expect from Swinburne in order to enjoy a quality learning experience. The Charter also sets out what is expected of students with regards to your studies and the way you conduct yourself towards other people and property.

You are expected to familiarise yourself with University regulations and policies and are obliged to abide by these, including the [Student Academic Misconduct Regulations](#), [Student General Misconduct Regulations](#) and the [People, Culture and Integrity Policy](#). Any student found to be in breach of these may be subject to disciplinary processes.

Examples of expected behaviours are:

- conducting yourself in teaching areas in a manner that is professional and not disruptive to others
- following specific safety procedures in Swinburne laboratories, such as wearing appropriate footwear and safety equipment, not acting in a manner which is dangerous or disruptive (e.g. playing computer games), and not bringing in food or drink
- following emergency and evacuation procedures and following instructions given by staff/wardens in an emergency response.

Canvas

You should regularly log on to the Swinburne learning management system, Canvas. You can access Canvas via the [Student login](#) webpage or <https://swinburne.instructure.com/> Canvas is updated regularly with important unit information and communications.

Communication

All communication will be via your Swinburne email address. If you access your email through a provider other than Swinburne, then it is your responsibility to ensure that your Swinburne email is redirected to your private email address.

Academic Integrity

Academic integrity is about taking responsibility for your learning and submitting work that is honestly your own. It means acknowledging the ideas, contributions and work of others; referencing your sources and acknowledging the use of generative artificial intelligence;

contributing fairly to group work; and completing tasks, tests and exams without cheating. Artificial intelligence tools should only be used where approved by the Unit Convenor.

Swinburne University uses the Turnitin system, which helps to identify inadequate citations, poor paraphrasing and unoriginal work in assignments that are submitted via Canvas. Your Unit Convenor will provide further details.

Plagiarism, collusion, contract cheating, unauthorised file sharing, falsification, fabrication, manipulation or misrepresentation of information, reuse of previous work and non-compliance with instructions in an invigilated or non-invigilated assessment item are all breaches of academic integrity and treated as academic misconduct. Examples of breaches of academic integrity include, but are not limited to:

- submitting work as your own for assessment that has been fully or partially completed by a third party, either paid or unpaid
- using output from artificial intelligence tools (e.g. ChatGPT) in whole or part without acknowledgement and/or without the approval of the Unit Convenor
- using another person's work or ideas as though it is your own work, without appropriate attribution
- working closely with another student or group of students (either past or current), to submit for assessment, some or all of the other student or students' work as your own work
- sharing without permission of the Unit Convenor, Swinburne resources or other material related to assessment to an entity or document repository site
- creating, intentionally modifying or inventing information that is intended to be submitted as part of an assessment item
- using the whole or part of a computer program written by another person as your own without appropriate acknowledgement
- poorly paraphrasing somebody else's work
- using a musical composition or audio, visual, graphic and photographic work created by another person without acknowledgment
- enabling others to cheat, including letting another student copy your work or by giving access to a draft or completed assignment
- letting someone or something else impersonate you, or you impersonate someone else in an invigilated or non-invigilated assessment item
- accessing, obtaining and/or providing to others unauthorised materials relating to an invigilated or non-invigilated assessment item.

The penalties for academic misconduct can be severe, ranging from a zero grade for an assessment task through to exclusion from Swinburne. For further details, see

<https://www.swinburne.edu.au/student-login/academic-integrity/>

Student support

Swinburne offers a range of services and resources to help you complete your studies successfully. Your Unit Convenor or studentHQ can provide information about the study support and other services available for Swinburne students. For further information, see the [Current students](#) web page.

Special consideration

If your studies have been adversely affected due to serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you may be able to apply for special consideration (SPC).

Applications for Special Consideration are submitted via the SPC online tool normally no later than 5.00pm on the third working day after the submission/sitting date for the relevant assessment component. See <https://www.swinburne.edu.au/life-at-swinburne/student-support-services/special-consideration-assistance/>

Note: Submitting fraudulent (fake or altered) medical certificates is considered misconduct and can lead to serious penalties from Swinburne. In addition, your doctor may report fraudulent medical certificates as a prosecutable offence under the Victorian Crimes Act.

AccessAbility Services

If you are a student with a disability, medical or mental health condition or you have significant carer responsibilities, you may require reasonable adjustments to fully access and participate in education. Swinburne's AccessAbility Services can develop an Education Access Plan (EAP) that includes the services and reasonable adjustments that you need.

It is recommended that you register with AccessAbility Services when you first commence your course but you can contact the service at any time during your studies to find out about reasonable adjustments. Contact [Accessability Services](#) to discuss further.

Review of marks

An independent marker reviews all fail grades for major assessment tasks. In addition, a review of assessment is undertaken if your final result is between 45 and 49 or within 2 marks of any grade threshold.

You can ask the Unit Convenor to check the result for an assessment item or your final result. Your request must be made in writing within 10 working days of receiving the result. The Unit Convenor can discuss the marking criteria with you and check the aggregate marks of assessment components to identify if an error has been made. This is known as local resolution. If you are dissatisfied with the outcome of the local resolution, you can lodge a formal complaint.

Feedback, complaints and suggestions

In the first instance, discuss any issues with your Unit Convenor. If your concerns are not resolved or you would prefer not to deal with your Unit Convenor, then you can complete a feedback form. See <https://www.swinburne.edu.au/corporate/feedback/>

Advocacy

If you require assistance with any academic issues, University statutes, regulations, policies and procedures, you are advised to seek advice from an Independent Advocacy Officer at the Swinburne Student Association. Talking to an Advocacy Officer is free, independent and confidential. For more information and booking an appointment, please see <https://www.swinburne.edu.au/current-students/student-services-support/advocacy/>